<u>SQL Injection Detection Using Machine Learning</u>

Student name: Mahmoud Fathy Nabil

Student id: 24102424

course name: Math for computing project

2. Abstract

This project aims to develop a machine learning model capable of detecting SQL injection attacks in web applications. SQL injections are one of the most common and dangerous web vulnerabilities. The proposed system uses text processing techniques and classification algorithms to distinguish between normal SQL queries and malicious ones. A dataset containing labeled SQL queries was used to train and evaluate the model. The Logistic Regression algorithm achieved high accuracy and demonstrated strong performance in detecting malicious inputs. The project also includes a simple web-based interface for real-time prediction.

3. Introduction

Web applications are increasingly targeted by cyberattacks, with SQL injection being one of the most prevalent threats. Attackers exploit vulnerabilities in input fields to execute unauthorized database queries.

Traditional security methods such as firewalls are not always sufficient. Therefore, machine learning offers a powerful alternative by learning patterns of malicious behavior and detecting them automatically.

This project focuses on building a machine learning-based system to classify SQL queries into normal or malicious categories.

4. Problem Statement

The main objective of this project is to design a system that can automatically detect SQL injection attacks from user input queries.

The challenge lies in distinguishing between legitimate SQL queries and malicious ones that may appear similar in structure.

5. Literature Review

Several studies have explored machine learning techniques for detecting SQL injection attacks.

- Research shows that **text-based classification models** can effectively identify malicious patterns.
- Techniques such as **TF-IDF vectorization** and **Logistic Regression** are commonly used.
- Deep learning methods like LSTM have also been proposed but require more computational resources.

In this project, we focus on a simpler and efficient approach suitable for practical implementation.

6. Dataset Description

The dataset used contains SQL queries labeled as:

- **0 → Normal Query**
- **1 → SQL Injection Attack**

Dataset Features:

- Query: Text of SQL statement
- Label: Classification (0 or 1)

The dataset includes thousands of examples, making it suitable for training a machine learning model.

7. Data Preprocessing

The following preprocessing steps were applied:

- Removal of missing values
- Text extraction from query column
- Label separation
- Conversion of text into numerical format using TF-IDF

TF-IDF:

Transforms text into numerical vectors based on word importance

8. Methodology

Steps followed:

1. Load dataset
2. Clean data
3. Convert text to features (TF-IDF)
4. Split data (80% training / 20% testing)
5. Train model
6. Evaluate performance

9. Model Implementation

Model 1: Logistic Regression

- Simple and efficient for text classification
- Achieved high accuracy

Model 2: Random Forest

- Tested for comparison
- Performed lower than Logistic Regression

10. Results & Evaluation

✅ Logistic Regression Results:

- Accuracy: **95%**
- Precision (Attack): High
- Recall (Attack): Good

Confusion Matrix Analysis:

- Correct predictions for normal queries: High
- Some attack queries were missing

✅ Random Forest Results:

- Accuracy: ~81%
- Less effective text data

11. Discussion

The results show that Logistic Regression is more suitable for this problem due to:

- Simplicity
- Efficiency with TF-IDF features
- Better generalization

However, the model still misses some attack queries, which can be improved in future work.

12. Conclusion

This project successfully demonstrates the use of machine learning in detecting SQL injection attacks.

The Logistic Regression model achieved high accuracy and proved to be effective for this task. The addition of a simple user interface allows real-time detection.

13. Future Work

- Use deep learning models (LSTM, CNN)
- Improve recall for attack detection
- Deploy system as a real web application
- Use larger and more diverse datasets

14. Demo Description

A simple web interface was developed using Streamlit where users can:

- Enter SQL queries
- Get instant prediction (Normal / Attack)

15. References (IEEE Style)

1. K. R. Chowdhary et al., "SQL Injection Detection Using Machine Learning," IEEE Access, 2020.
2. M. Kamtuo et al., "Detecting SQL Injection Using Machine Learning," 2021.
3. Scikit-learn Documentation.
4. Kaggle SQL Injection Dataset.